Nebula Business Operations Hub

Project Review Guide

Modules covered:

Recruitment / ATS | Expenses | Inventory | HRMS

> **Purpose:** A review-ready explanation of the complete business flow, module responsibilities, database lifecycle, approvals, AI features, cross-module integrations, candidate resume location, and Recruitment Talent Pool.

Technology: React + TypeScript + TanStack Query + Axios, Node.js + Express, Mongoose + MongoDB, JWT authentication, RBAC and organization-level tenant isolation.

1. Common System Architecture

```
React frontend
  → TanStack Query state/cache
  → Axios authenticated request
  → Express authentication and permission middleware
  → organizationId tenant scoping
  → controller and service business logic
  → Mongoose
  → MongoDB
  → response and UI refresh
```

- **Authentication:** JWT identifies the current user and organization.
- **Multi-tenancy:** Queries are scoped by `organizationId`, preventing cross-company data access.
- **RBAC:** Actions such as view, create, edit, delete and approve are permission controlled.
- **Auditability:** Important transitions create immutable or chronological activity records.

2. Recruitment / ATS Module

The Recruitment module manages the complete journey from identifying a vacancy to converting a selected candidate into an HRMS employee.

```
Manpower requirement → job requisition/opening → AI advertisement → candidate
application
→ screening pipeline → interviews and scorecards → offer approval → acceptance
→ one-click HRMS onboarding
```

Step-by-step flow

1. **Create job opening:** HR or a hiring manager enters title, department, employment type, location, experience, salary band, skills, recruiters, hiring manager and open positions. The normal status flow is `draft → published → on_hold → closed`.
2. **Generate advertisement:** Gemini generates a job post in dynamic, technical or corporate tone. A deterministic template fallback keeps the feature usable if AI is unavailable.
3. **Receive candidate:** Candidate profile stores contact details, resume, experience, compensation expectation, notice period, LinkedIn/GitHub/portfolio and source. A code such as `CAND-2026-0012` is generated.
4. **Manage pipeline:** Recruiters use Kanban stages such as Applied, Screening, Interview R1/R2, Technical Assessment, HR Round, Offer Sent and Hired. Rejected and Withdrawn are terminal states. Stage changes are audited.
5. **Schedule interviews:** Store round type, interview panel, start/end time, time zone, meeting link or room and status. Status can become completed, rescheduled, cancelled or no-show.
6. **Evaluate candidate:** Interviewers submit 1–5 ratings for technical ability, communication and culture fit, plus notes and a hire/hold/reject recommendation.

7. **Create offer:** HR enters designation, department, compensation, currency, joining date, benefits and validity. State flow: `draft → pending_approval → approved → sent → accepted`.
8. **Onboard to HRMS:** The bridge validates the candidate, creates a unique Employee record, migrates resume and offer letter to EmployeeDocument, marks the candidate hired, increments filled positions, updates offer status and logs the conversion.

Where is the candidate resume?

> **Recruitment → ATS Pipeline/Candidates → click candidate card → Candidate Drawer → Candidate Details → Resume Viewer → View/Preview/Download.**

The candidate profile references the uploaded file through `resumeUrl`. After hiring, the same resume is available in the employee's HRMS document repository.

Talent Pool: meaning and flow

The **Talent Pool** is a reusable collection of candidates who may be suitable for current or future vacancies. It prevents good candidates from being lost when a job closes, a candidate is not selected for one role, or no matching job is currently open.

```
Candidate sourced/applied → profile and resume saved → skills/tags/search data maintained
→ candidate retained in pool → recruiter searches future opening → candidate linked/advanced to pipeline
→ interview/offer/hire or remains available
```

- It can contain passive candidates, referrals, previous applicants, silver-medalist candidates and agency-sourced profiles.
- Recruiters can search using skills, experience, location, notice period, source, previous stage and job compatibility.
- Talent Pool is not the same as the active pipeline: the pipeline is tied to a particular job; the pool is reusable across jobs.
- A candidate can move from the pool into a specific job's Applied or Screening stage without recreating the profile.

3. Expenses Module

The Expenses module controls employee business spending from receipt capture through two-level approval, reimbursement and accounting export.

```
Expense incurred → receipt upload/AI scan → multi-item claim → policy and budget validation
→ employee submission → manager approval → finance approval → reimbursement → ERP CSV export
```

Step-by-step flow

1. **Create claim:** An employee bundles multiple items such as travel, hotel, meals or taxi. Each item stores category, date, merchant, description, amount, tax, currency, receipt and billable-to-client information.
2. **Scan receipt:** Gemini Vision extracts merchant, date, amount, tax, currency, description and suggested category. The employee verifies the result; a fallback parser is available when AI is unavailable.
3. **Apply policy:** Category rules check maximum claim limits, receipt thresholds, active status and GL code mapping.
4. **Check budget:** The system calculates spent amount, pending amount, remaining amount and percentage used. Less than 80% is normal, 80–99% is warning and above 100% is a breach.
5. **Submit:** Claim status changes from `draft` to `submitted`; a number such as `EXP-2026-0012` and an activity record are created.
6. **Manager review:** The line manager checks business purpose, receipt and departmental ownership. Approval gives `manager_approved`; rejection requires a reason. Self-approval is blocked.
7. **Finance review:** Finance checks receipt, GL code, tax, budget and policy compliance. Approval gives `approved`.
8. **Reimburse and export:** Finance records bank transfer, payroll, corporate card, cash or cheque, including transaction reference. The final state is `reimbursed`; CSV export supports ERP reconciliation.

4. Inventory Module

Inventory manages item masters, warehouses, procurement, receiving, stock valuation, traceability, equipment assets and supply-chain analytics.

```
Item/warehouse setup → Purchase Requisition → budget check → Purchase Order
→ Goods Receipt Note → immutable Stock Ledger → FIFO/WAC valuation
→ issue/transfer/asset assignment → analytics and accounting events
```

Step-by-step flow

1. **Configure master data:** Define item SKU, category, UoM, barcode, reorder level and tracking type. Configure Warehouse → Zone → Bin and supplier terms.

2. **Create PR:** A department requests material. The system checks budget through Expenses and prevents self-approval.
3. **Create PO:** An approved PR becomes a supplier Purchase Order with prices, taxes, terms and versioned amendments.
4. **Receive with GRN:** Warehouse staff record ordered, received, accepted and rejected quantities, plus batches or serial numbers and inspection results.
5. **Post ledger:** Accepted receipt adds a positive StockLedger entry. Issues add negative entries. Corrections use compensating reversals rather than editing history.
6. **Protect stock:** Available stock equals on-hand minus reserved stock. An issue exceeding available quantity is rejected before posting.
7. **Value stock:** FIFO consumes the oldest cost layer first; Moving WAC recalculates the average cost after each receipt.
8. **Transfer and trace:** Inter-warehouse movement follows dispatch → in_transit → receipt. Batch and serial lineage supports forward and backward traceability.
9. **Three-way match:** The system compares Purchase Order, GRN and supplier invoice quantities and prices. Out-of-tolerance differences can block payment or create a debit note.
10. **Analyze:** Forecasting, OCR, anomaly radar, ABC/XYZ matrix, aging, turnover, DSI and supplier OTIF reports support decisions.

5. HRMS Module

HRMS manages employees, organization hierarchy, attendance, shifts, leave, payroll, documents, performance and workforce intelligence.

```
User/Employee identity → organization and job profile → shift assignment → attendance
→ leave and accruals → payroll → payslip/payment → performance and career → offboarding
```

Employee and identity flow

The system keeps authentication identity in `User` and personnel data in `Employee`, linked through canonical identity resolution. Employee profiles include department, designation, branch, manager, statutory details, bank/payroll information, documents and job history.

Attendance and shifts

1. HR configures morning, evening, night or custom shifts with grace and half-day rules.
2. HR assigns a shift to the employee.
3. The employee performs clock-in, break-in, break-out and clock-out.
4. The system calculates worked minutes, lateness and attendance status.
5. A missed punch can be submitted for manager regularization and automatic reconciliation.
6. Weekend or holiday work may generate a compensatory-off credit.

Leave flow

1. HR defines leave types, quotas, accrual, carry-forward and caps.
2. Employee submits a request with dates and reason.
3. The system previews sandwich-policy days where weekends or holidays fall between leave days.
4. Manager approves or rejects. Approval debits the LeaveLedger; rejection restores any reserved balance.
5. A monthly accrual process adds credits and records each transaction.

Payroll flow

```
Salary structure + attendance + approved leave + holidays + LOP
→ earnings and deductions → PayrollRun → Payslip → approval → paid
```

- Salary structures define Basic, HRA, allowances, PF, PT, TDS and ESI.
- Payroll reads attendance and approved leave to calculate loss-of-pay days.
- Monthly payroll moves through `draft → approved → paid`.
- PDF payslips contain attendance, earnings, deductions and net salary and are available in the employee self-service vault.

Performance and talent

Employees and managers use weighted OKRs, milestones, check-ins, 360-degree review cycles and the 3×3 performance-potential matrix. Attrition analysis uses attendance, leave, tenure and review trends to identify retention risk.

6. Cross-Module Integrations

Integration	Flow	Business value
Recruitment → HRMS	Accepted candidate becomes Employee; resume and offer letter become EmployeeDocuments.	Zero duplicate onboarding entry.
HRMS → Inventory	New employee role selects starter kit; assets are reserved and serial numbers assigned.	Controlled equipment provisioning and recovery.
Inventory → Expenses	Purchase Requisition checks department budget before approval.	Prevents unauthorized departmental overspending.
HRMS → Expenses	Employee, department and manager determine claim ownership and approval route.	Correct reimbursement routing.
HRMS → Payroll	Attendance, approved leave and salary structure produce payroll and payslips.	Attendance-driven salary calculation.

7. Complete Review Example

1. Engineering creates a Software Engineer vacancy and publishes an AI-assisted job post.
2. A candidate applies, enters the ATS pipeline, attends interviews and receives scorecards.
3. HR approves and sends an offer; the candidate accepts.
4. The HRMS bridge creates the employee and migrates the resume and offer letter.
5. Inventory selects a Software Engineer kit and assigns a serial-numbered laptop.
6. The employee punches attendance, applies for leave and is included in monthly payroll.
7. The employee scans a client-meal receipt, submits an expense claim, receives manager and finance approval, and is reimbursed.

> **Final review statement:** Nebula is not a set of isolated CRUD screens. It is an integrated, multi-tenant ERP workflow in which recruitment creates the workforce, HRMS governs the workforce, inventory equips the workforce, and expenses controls business spending.

8. Quick Viva Answers

- **What is the Talent Pool?** A reusable database of potential candidates available for current or future vacancies, separate from a job-specific active pipeline.
- **Where is the resume?** Recruitment → ATS Pipeline/Candidates → Candidate Card → Candidate Drawer → Candidate Details → Resume Viewer.
- **Why use an immutable inventory ledger?** To preserve audit history and correct mistakes through linked reversals instead of silently changing past movements.

- **Why two expense approvals?** The manager validates business purpose; Finance validates policy, accounting and payment.
- **What is the strongest integration?** The Recruitment-to-HRMS bridge, because one accepted candidate becomes a complete employee without re-entering data.